Development summary: stick-knot counting

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Discussions and model prompting: Alexander Kolpakov (UATX). This retrospective account summarizes the released manuscript and its supporting mathematical documentation. It is not a transcript or a reconstruction of model turns. No original conversation export was available in the source tree.

Schematic research brief

The task is to determine the growth of the number of ordinary knot types realizable with a budget of at most N straight sticks. A schematic research brief is to find an upper bound from polygon geometry, a complementary lower construction, and a precise account of the mathematical inputs. This is an editorial description of the task, not a quotation of an original prompt.

For the set $\mathcal{K}_N$ of these knot types, the manuscript presents

$$N^{(2/3+o(1))N} \leq |\mathcal{K}_N| \leq N^{3N+o(N)}, \qquad \log|\mathcal{K}_N| = \Theta(N \log N).$$

This determines the logarithmic growth order. It does not determine a leading constant or an asymptotic equivalent for the number of knot types.

Mathematical development

1. **Turn knot geometry into a low-degree wall problem.** Describe an N-vertex polygon by $3N$ real coordinates. Cubic determinants detect the possible intersection of nonincident edges, while quartic squared cross products exclude collapsed or overlapping adjacent edges. The number of walls grows only quadratically in N.

2. **Pass from chambers to knot types.** Every bounded-stick knot has a generic representative in the wall-free open locus. Knot type is constant along paths, so a surjection from connected components gives an upper bound for the knot population.

3. **Keep the factorial saving in the component count.** Apply the refined real-algebraic component estimate with its partial binomial sum retained. The dimension factorial changes the relevant base from quadratic to linear in N, producing the upper exponent 3.

4. **Build a large family of distinct braid words.** Four independent permutations on k letters give at least $(k!)^4$ doubly alternating permutations on $4k$ letters. Starting and finishing conditions make the corresponding words distinct normal forms.

5. **Control the endpoint and stick costs.** Pigeonhole by endpoint permutation and append a common correcting factor to obtain distinct pure braids with controlled length. Transfer these to knots through the cited braid-to-knot and arc-to-stick results, including the bounded loss on forgetting orientation.

6. **Optimize and check the counting deductions.** Suitable fixed word lengths give every lower coefficient below $2/3$. A logarithmically varying word length yields the refined lower deficit $-(2/3)N \log\log N - O(N)$. The resulting upper and lower estimates give the common $N \log N$ scale for the logarithm.

The upper argument counts discriminant chambers directly. It is independent of related strong-geometry reconstruction results. The manuscript states its priority claim with the qualification "to our knowledge" and keeps the original real-algebraic, braid, and knot-theoretic inputs attributed.

Verification and scope

The standalone Lean project uses actual polygon coordinates, wall polynomials, paths and components, permutation words, endpoint fibers, and a bounded-stick subtype of a supplied knot interface. It checks the internal finite and asymptotic deductions, including the optimized floor/log estimate.

Ambient-isotopy invariance, the Barone–Basu component theorem, structural Garside facts, the pure-braid-to-knot injection, the arc-to-stick inequality, and orientation reversal enter through explicit external mathematical hypotheses. The classical ambient-isotopy quotient and the auxiliary link construction are outside the formal target. The formal certificate is conditional on those stated inputs.

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